(4.) Network Function Virtualization (NFV)

1. Was ist Network Softwarization? Welche Ausprägungen gibt es und was ist deren Basis?

- Network Softwarization is the transition from hardware-based network function to software running on general-purposed hardware. Its main forms are SDN and NFV, both based on virtualization and programmability.

2. Wie können Multi-Tenant-Netze in virtualisierten Umgebungen realisiert werden? Motivation und Herausforderungen?

- They are realized using technologies like VLANs or VXLANs overlays to isolate traffic between different customers on shared hardware. The motivation is cost-sharing through resource pooling; the challenge is ensuring strict security and performance isolation.

3. Wie werden virtuelle Switches (vSwitches) realisiert? Welche Funktionen werden erbracht?

- vSwitches are software implementations (like Open vSwitch) running within a Hypervisor. They bridge traffic between virtual Machines (VM) and the physical network, supporting features like packet filtering, tagging, and monitoring.

4. Beschreiben Sie den Einsatz mehrerer Netzwerkkarten bei der Server-Virtualisierung (Ausfallsicherheit / Lastverteilung).

- Multiple NICs are used for NIC Teaming or Bonding. If one card fails, traffic is instantly rerouted to the other (failover), and during normal operation, they can combine their bandwidth to balance the traffic load.